

Project Executable Files

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA
6	Deployed application URL (if hosted)	No
7	APK / Executable binary (if applicable for mobile/desktop apps)	Yes
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	No

Step 2: File / Folder Structure

OptiCrop/

```

|
|— static/
|   |— style.css
|
|— templates/
|   |— index.html
|   |— result.html
|
|— app.py
|— train_model.py
|— model.pkl
|— requirements.txt
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Run locally on loopback address: http://127.0.0.1:5000/
Login Credentials (Demo)	Local Host Server Environment (WSGI Development Server Layer)
Platform / Hosting Provider	Open Session Access — No authentication middleware barrier required for evaluation.
Repository Link	https://github.com/HariCharan-1912/OptiCrop-Smart-Agricultural-Production-Optimization-Engine.git
Demo Video Link	https://drive.google.com/drive/folders/1m7iTh-894ZB1jqWrpKp90yuZzstdy0bB?usp=drive_link

Step 4: Run Instructions:

1. Open Workspace: Open the root OptiCrop folder inside **VS Code**.

2. Activate Environment: Open the built-in terminal (Ctrl + ~) and activate your conda base path to ensure package isolation:

```
Bash: C:\Users\haric\anaconda3\Scripts\activate
```

3. Install Dependencies: Cleanly download the pre-compiled Python binaries using the no-cache directive:

```
Bash: python -m pip install --no-cache-dir -r requirements.txt
```

4. Compile the Machine Learning Pipeline: Execute the trainer module to evaluate the classifiers and output the active model weights:

```
Bash: python train_model.py
```

Verification: Confirm that model.pkl appears inside your file directory tree sidebar.

5. Launch the Engine Web Core: Start up the server gateway routing daemon:

```
Bash: python app.py
```

6. Access Dashboard: Copy and paste the address <http://127.0.0.1:5000/> into any modern internet browser to test all three crop optimization scenarios.